Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

High-Performance Computing Lab for CSE **2026**

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Report for Project 1

Due date: 09 March 2026, 18:00 (Europe/Zurich)

HPC Lab for CSE 2025 — Submission Instructions

Following these instructions is mandatory: Submissions that do not comply **will not be considered**.

- **Submission Platform**: Assignments must be submitted via SAM-UP.
- **Required Files**: Your submission must include:
  - **Report**: A PDF summarizing your results and observations, based on this LaTeX template.
    - File name: `project_number_lastname_firstname.pdf`
    - You are allowed to discuss all questions with anyone you like; however: (i) your submission must list anyone you discussed problems with and (ii) you must write up your submission independently.
  - **Source Code Archive**: A **compressed tar archive**, named `project_number_lastname_firstname.tar.gz`, containing:
    - All source code files.
    - Build files and scripts (e.g., `Makefile`, batch job scripts, plot scripts, etc). If you modified any provided files, ensure they still work as expected.
    - The full LaTeX source of your report including all needed files to build the report (figures, listings, etc).
  - **Size Limit**: The total submission (PDF + archive) must not exceed **25 MB**.
  - **File Naming Rules**: Use only `ASCII` characters (avoid spaces, special symbols, and non-English letters).
- **Grading Process**: The TAs will evaluate your project based on:
  - Your report write-up.
  - Your code implementation.
  - Benchmarking your code's performance.

1. Euler warm-up [10 points]

Question 1: What is the module system, and how is it used?
The module system (Lmod, a Lua-based environment modules implementation) is used on HPC systems to dynamically manage software environments. It allows users to load, unload, and switch between different software versions without manually modifying system paths.
Loading all software globally would lead to conflicts between different versions of compilers and libraries. The module system avoids this problem by dynamically modifying environment variables.
In practice, before compiling or running programs on Euler, users load the required modules (e.g., compilers or libraries) inside their batch scripts using commands such as `module load`.

Question 2: What is Slurm, and what is its purpose?
Slurm is an open-source workload manager and job scheduler for Linux and Unix-like systems, widely used by supercomputers and clusters. It schedules and executes jobs on compute nodes, handling resource allocation, queuing, and execution control.
In particular (from the Slurm documentation), it provides three main functions:

- it allocates exclusive and/or non-exclusive access to resources (compute nodes, CPUs, memory, GPUs) to users for a specified duration of time so they can perform work;
- it provides a framework for starting, executing, and monitoring work (normally a parallel job) on the set of allocated nodes.
- it arbitrates contention for resources by managing a queue of pending jobs.

We now illustrate how the remaning questions were adressed.
The solution to question 3 and 4 can be found by first running the Makefile provided in the directory `1_Euler_Warmup` and then running the file `slurm_job_one.sh`. We already provide in the directory `1_Euler_Warmup/build_job_one` the ouputs of the batch script as requested. The output is reported also here for completeness:

```
Currently Loaded Modules:
1) stack/2025-06 2) gcc/12.2.0



Model name: AMD EPYC 9654 96-Core Processor
eu-g9-101-4
Hello World from eu-g9-101-4
```

As we can see we have specified to use a node with AMD EPYC 9654 96-Core Processor, and the job was executed on the node `eu-g9-101-4`.
The solution to question 5 can be found by running the file `slurm_job_two.sh`. The output of the batch script is reported here for completeness:

```
Currently Loaded Modules:
  1) stack/2025-06 2) gcc/12.2.0



Model name: AMD EPYC 7763 64-Core Processor
Model name: AMD EPYC 7763 64-Core Processor
eu-a2p-487
eu-a2p-482
Hello World from eu-a2p-487
Hello World from eu-a2p-482
```

We have thus successfully executed the same job on two nodes, both with AMD EPYC 7763 64-Core Processor, and jobs were executed on the nodes `eu-a2p-487` and `eu-a2p-482` respectively.

2. Performance characteristics [50 points]

2.1. Peak performance [16 Points]

Proceeding as in the example given in the project description, the hardware specifications can be obtained from the Euler webpage.
For Euler VII (phase 1), the system consists of $n_{\text{nodes}} = 292$ nodes, each equipped with $n_{\text{sockets}} = 2$ sockets and $n_{\text{cores}} = 64$ cores per socket, operating at a frequency $f = 2.6\,\text{GHz}$.
For Euler VII (phase 2), the system consists of $n_{\text{nodes}} = 248$ nodes, each with $n_{\text{sockets}} = 2$ sockets and $n_{\text{cores}} = 64$ cores per socket, operating at a frequency $f = 2.45\,\text{GHz}$.

Using the uops.info database provided, we determine both the superscalability factor and the FMA factor for the corresponding AMD architectures.
For both Euler VII phases, which are based on AMD Zen2 and AMD Zen3 respectively, we obtain $n_{\text{super}} = 4$ and $n_{\text{FMA}} = 2$.

The FMA factor is determined by observing that all instructions marked as FMA in the table have a corresponding number of micro-operations equal to one. This indicates that the fused multiply-add is executed as a single micro-operation by dedicated hardware. Since an FMA instruction performs the operation $a \cdot b + c$, it consists of one floating-point multiplication and one floating-point addition. Therefore, each FMA instruction corresponds to two floating-point operations, and thus $n_{\text{FMA}} = 2$.

The superscalability is defined as the number of floating-point operations that the FPU of a core can execute in parallel within a single clock cycle. To determine this value, we analyze the instruction `VFMADD132PD (YMM, YMM, YMM)` reported in the uops.info table. The table specifies a throughput of 0.50 cycles per instruction, one micro-operation, and execution on ports FP0/1. A throughput of 0.50 cycles per instruction implies that

$$\frac{1}{0.50} = 2$$

FMA instructions can be issued per cycle. The port information FP0/1 confirms the presence of two independent floating-point execution pipelines capable of executing FMA instructions in parallel.

Since two FMA instructions can be executed per cycle and each FMA instruction corresponds to two floating-point operations, the total number of floating-point operations per cycle is

$$2 \times 2 = 4.$$

Hence, for both AMD Zen 2 and AMD Zen 3, the superscalability is 4 floating-point operations per cycle per core.

To determine n_{SIMD}, we inspect the floating-point instructions reported in the uops.info table, where instructions such as `VFMADD132PD (YMM, YMM, YMM)` appear. The presence of YMM operands indicates the use of 256-bit AVX registers, since YMM registers are defined as 256-bit wide in the AVX/AVX2 instruction set. As both AMD Zen 2 and Zen 3 execute these YMM-based instructions, they necessarily support AVX2 with 256-bit SIMD registers. A double-precision floating-point number occupies 64 bits; therefore, the number of doubles that can be processed concurrently by one SIMD instruction is

$$n_{\text{SIMD}} = \frac{256}{64} = 4.$$

Hence, each SIMD instruction can operate on four double-precision values in parallel.

Parameter	**Euler VII (Phase 1)**	**Euler VII (Phase 2)**
n_{nodes}	292	248
n_{sockets}	2	2
n_{cores} (per socket)	64	64
Clock frequency f	2.6 GHz	2.45 GHz
n_{FMA}	2	2
n_{super} (FP ops/cycle/core)	4	4
n_{SIMD} (doubles per SIMD instr.)	4	4

Table 1: Summary of hardware parameters and architectural factors used for peak performance estimation.

We can now compute all the peak performance parameters for both Euler VII phases :

$$P_{\text{core},1} = n_{\text{super}} \times n_{\text{FMA}} \times n_{\text{SIMD}} \times f = 4 \times 2 \times 4 \times 2.6\,\text{GHz} = 83.2\,\text{GFLOPS/core}.$$
$$P_{\text{CPU},1} = n_{\text{cores}} \times P_{\text{core},1} = 64 \times 83.2\,\text{GFLOPS/core} = 5.3248\,\text{TFLOPS/CPU}.$$
$$P_{\text{node},1} = n_{\text{sockets}} \times P_{\text{CPU},1} = 2 \times 5.3248\,\text{TFLOPS/node} = 10.6496\,\text{TFLOPS/node}.$$
$$P_{\text{cluster},1} = n_{\text{nodes}} \times P_{\text{node},1} = 292 \times 10.6496\,\text{TFLOPS/node} = 3110.08\,\text{TFLOPS}.$$

$$P_{\text{core},2} = n_{\text{super}} \times n_{\text{FMA}} \times n_{\text{SIMD}} \times f = 4 \times 2 \times 4 \times 2.45\,\text{GHz} = 78.4\,\text{GFLOPS/core}.$$
$$P_{\text{CPU},2} = n_{\text{cores}} \times P_{\text{core},2} = 64 \times 78.4\,\text{GFLOPS/core} = 5.017\,\text{TFLOPS/CPU}.$$
$$P_{\text{node},2} = n_{\text{sockets}} \times P_{\text{CPU},2} = 2 \times 5.017\,\text{TFLOPS/node} = 10.034\,\text{TFLOPS/node}.$$
$$P_{\text{cluster},2} = n_{\text{nodes}} \times P_{\text{node},2} = 248 \times 10.034\,\text{TFLOPS/node} = 2488.432\,\text{TFLOPS}.$$

2.2. Memory Hierarchies [8 Points]

2.2.1. Cache and main memory size

The solution to this exercise can be found by first running the Makefile in `2_Performance_Characteristics/Memory_Hierarchies` and then executing `memory_hierarchies_01.sh` for Euler VII (phase 1). The batch-script output is available in `2_Performance_Characteristics/Memory_Hierarchies/build_1`.
We report in table 2 the results for Euler VII (phase 1) and in figure 1 the memory topology.

Main memory (total)	249 GB
L3 cache (total)	512 MB (32 × 16 MB)
L3 cache (per socket)	256 MB (16 × 16 MB)
L2 cache (per core)	512 KB
L1 data cache (per core)	32 KB
L1 instruction cache (per core)	32 KB

Table 2: Memory hierarchy of an Euler compute node with AMD EPYC 7H12 (2 sockets, 64 cores per socket). Each core has private L1 and L2 caches, while L3 cache is organized in 32 blocks of 16 MB shared among groups of 4 cores.

We proceed in the same way for Euler VII (phase 2) by executing `memory_hierarchies_02.sh` and we report in table 3 the results for Euler VII (phase 2) and in figure 2 the memory topology.

Main memory (total)	249 GB
L3 cache (total)	512 MB (16 × 32 MB)
L3 cache (per socket)	256 MB (8 × 32 MB)
L2 cache (per core)	512 KB
L1 data cache (per core)	32 KB
L1 instruction cache (per core)	32 KB

Table 3: Memory hierarchy of an Euler compute node with AMD EPYC 7763 (2 sockets, 64 cores per socket). Each core has private L1 and L2 caches, while the L3 cache is organized in 16 blocks of 32 MB shared among groups of cores.

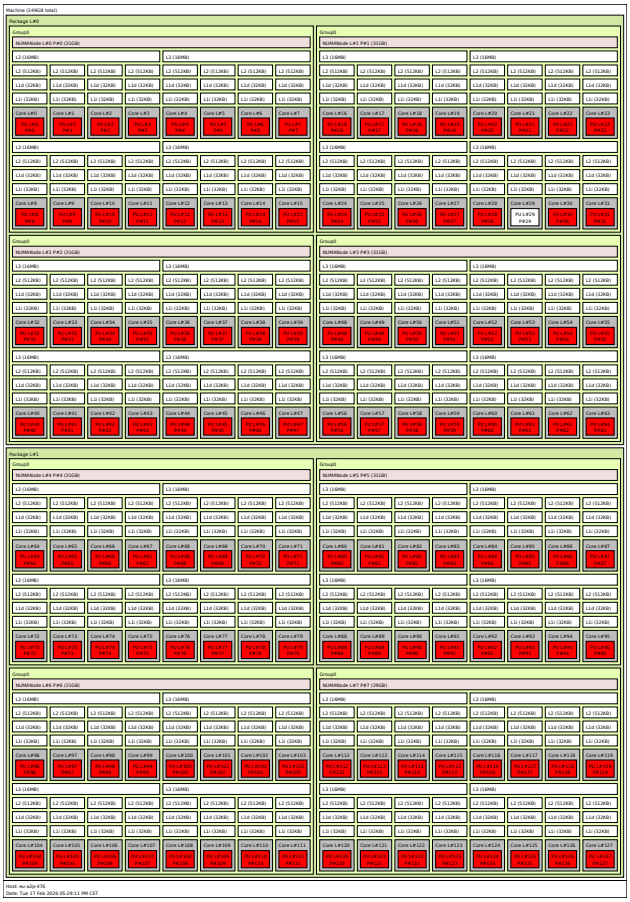


Figure 1: Topology of Euler VII (phase 1) compute node with AMD EPYC 7H12 (2 sockets, 64 cores per socket).

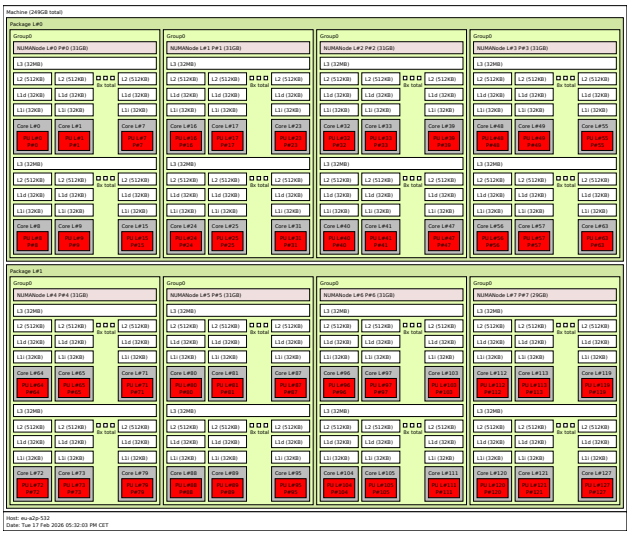


Figure 2: Topology of an Euler compute node with AMD EPYC 7763 (2 sockets, 64 cores per socket).

2.3. Bandwidth: STREAM benchmark [10 Points]

For this exercise one can run the Makefile in `2_Performance_Characteristics/STREAM_Benchmark` and then execute `stream_one.sh` for Euler VII (phase 1) and `stream_two.sh` for Euler VII (phase 2). The batch-script outputs are available in `2_Performance_Characteristics/STREAM_Benchmark/build_1` and `2_Performance_Characteristics/STREAM_Benchmark/build_2` respectively.

For Euler VII (phase 1), we run the file stream.c with the parameters specified in the following code

snippet, which corresponds to the command used in the batch script `stream_one.sh`:

```
gcc -O3 -march=native -mcmodel=medium -DSTREAM_TYPE=double -DSTREAM_ARRAY_SIZE=268435456
    -DNTIMES=20 stream.c -o build_1/stream_one_c.exe
```

We used as array size the value of 2^{28}, which corresponds to 2 GB per array, and we set the number of iterations to 20. The corresponding output of the batch script is reported here for completeness:

```
  Currently Loaded Modules:
  1) stack/2025-06 2) gcc/12.2.0




Model name: AMD EPYC 7H12 64-Core Processor
eu-a2p-476
-------------------------------------------------------------
STREAM version $Revision: 5.10 $
-------------------------------------------------------------
This system uses 8 bytes per array element.
-------------------------------------------------------------
 Array size = 268435456 (elements), Offset = 0 (elements)
Memory per array = 2048.0 MiB (= 2.0 GiB).
Total memory required = 6144.0 MiB (= 6.0 GiB).
Each kernel will be executed 20 times.
 The *best* time for each kernel (excluding the first iteration)
 will be used to compute the reported bandwidth.
-------------------------------------------------------------
Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 151041 microseconds.
   (= 151041 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.
-------------------------------------------------------------
WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.
-------------------------------------------------------------
Function Best Rate MB/s Avg time Min time Max time
Copy: 33537.1 0.133024 0.128066 0.146805
Scale: 18170.2 0.246928 0.236374 0.258205
Add: 20230.7 0.332126 0.318450 0.354032
Triad: 19668.1 0.341913 0.327558 0.364689
-------------------------------------------------------------
Solution Validates: avg error less than 1.000000e-13 on all three arrays
-------------------------------------------------------------
```

We choose `STREAM_ARRAY_SIZE` $= 2^{28}$ (268,435,456 doubles) so that each array occupies 2 GiB and the total working set (3 arrays $\approx$ 6 GiB) is well beyond the last-level cache. This reduces cache effects and makes the reported rates representative of sustained main-memory bandwidth. STREAM reports the *Best Rate* excluding the first iteration; for the roofline model we use the Triad kernel as a proxy for sustainable bandwidth, since it stresses both loads and stores and is commonly the most representative. The Copy kernel can appear faster because it benefits from efficient store handling (e.g., write combining), whereas Add/Triad are more load-intensive. Here, Triad achieves 19.7 GB/s (19,668 MB/s), which we use to define B_1 (after converting MB/s to GB/s). For Euler VII (phase 2), we run the file stream.c with the parameters specified in the following code snippet, which corresponds to the command used in the batch script `stream_two.sh`:

```
gcc -O3 -march=native -mcmodel=medium -DSTREAM_TYPE=double -DSTREAM_ARRAY_SIZE=20000000 -
    DNTIMES=20 stream.c -o build_2/stream_two_c.exe
```

Here instead we used as array size the value of 2×10^7, which corresponds to 160 MB per array, and we set the number of iterations to 20. The corresponding output of the batch script is reported here for completeness:

```
Currently Loaded Modules:
  1) stack/2025-06 2) gcc/12.2.0




Model name: AMD EPYC 7763 64-Core Processor
eu-a2p-527
-------------------------------------------------------------
STREAM version $Revision: 5.10 $
-------------------------------------------------------------
This system uses 8 bytes per array element.
-------------------------------------------------------------
Array size = 20000000 (elements), Offset = 0 (elements)
Memory per array = 152.6 MiB (= 0.1 GiB).
Total memory required = 457.8 MiB (= 0.4 GiB).
Each kernel will be executed 20 times.
 The *best* time for each kernel (excluding the first iteration)
 will be used to compute the reported bandwidth.
-------------------------------------------------------------
Your clock granularity/precision appears to be 1 microseconds.
Each test below will take on the order of 14061 microseconds.
   (= 14061 clock ticks)
Increase the size of the arrays if this shows that
you are not getting at least 20 clock ticks per test.
-------------------------------------------------------------
WARNING -- The above is only a rough guideline.
For best results, please be sure you know the
precision of your system timer.
-------------------------------------------------------------
Function Best Rate MB/s Avg time Min time Max time
Copy: 29744.9 0.011587 0.010758 0.012729
Scale: 19885.6 0.017333 0.016092 0.019567
Add: 15889.9 0.032855 0.030208 0.040448
Triad: 17459.6 0.031916 0.027492 0.044562
-------------------------------------------------------------
Solution Validates: avg error less than 1.000000e-13 on all three arrays
-------------------------------------------------------------
```

Note that `STREAM_ARRAY_SIZE` $= 2 \times 10^7$ with `STREAM_TYPE=double` corresponds to $2 \times 10^7 \cdot 8 \approx 160\,\text{MB}$ per array (as confirmed by STREAM: 152.6 MiB). Therefore the total working set is only $\approx 0.4\,\text{GiB}$ (three arrays), which is comparable to (or smaller than) the last-level cache and can lead to cache-influenced bandwidth numbers rather than sustained DRAM bandwidth. For a fair roofline bound, the array size should be increased so that the three arrays clearly exceed LLC (e.g., by a factor 4×), similarly to phase 1. Using the current run as-is, we still take the Triad kernel as a proxy for sustainable bandwidth (here $\approx 17.46\,\text{GB/s}$), but we interpret it cautiously because it may overestimate (or otherwise bias) the true main-memory bandwidth due to caching effects.

2.4. Performance model: A simple roofline model [16 Points]

A (naive) roofline model provides an upper bound on the attainable performance of a kernel as a function of its operational intensity I (in Flops/Byte). For a given hardware, the attainable performance is bounded by:

- a **compute ceiling** given by the peak floating-point performance P_{peak} (in GFlops/s),
- a **bandwidth ceiling** given by $B \cdot I$, where B is the sustainable memory bandwidth (in GBytes/s).

Hence the roofline bound is

$$P(I) = \min\left(P_{\text{peak}},\ B \cdot I\right). \tag{1}$$

The transition between memory-bound and compute-bound regimes occurs at the **ridge point**:

$$I_{\text{ridge}} = \frac{P_{\text{peak}}}{B}. \tag{2}$$

A kernel with $I < I_{\text{ridge}}$ is memory-bound, while a kernel with $I > I_{\text{ridge}}$ is compute-bound.

From the previous exercises, we have already computed the peak performance P_{peak} and the sustainable memory bandwidth B for both Euler VII phases. The values of the single core peak performances where

$$P_{\text{peak},1} = 83.2 \text{ GFlops/s} \quad \text{(Euler VII phase 1, AMD EPYC 7H12 @ 2.6 GHz)}, \tag{3}$$
$$P_{\text{peak},2} = 78.4 \text{ GFlops/s} \quad \text{(Euler VII phase 2, AMD EPYC 7763 @ 2.45 GHz)}. \tag{4}$$

We use the STREAM `Triad` bandwidth as sustainable memory bandwidth, since it represents a typical streaming workload (three arrays, two loads + one store per iteration) and is commonly used in roofline constructions. From our STREAM outputs, we have the following sustainable memory bandwidths:

$$B_1 = 19668.1 \text{ MB/s} = 19.6681 \text{ GB/s}, \tag{5}$$
$$B_2 = 17459.6 \text{ MB/s} = 17.4596 \text{ GB/s}, \tag{6}$$

where we convert MB/s to GB/s by dividing by 1000.
Using Eq. (2):

$$I_{\text{ridge},1} = \frac{P_{\text{peak},1}}{B_1} = \frac{83.2}{19.6681} \approx 4.23 \text{ Flops/Byte}, \tag{7}$$
$$I_{\text{ridge},2} = \frac{P_{\text{peak},2}}{B_2} = \frac{78.4}{17.4596} \approx 4.49 \text{ Flops/Byte}. \tag{8}$$

Quantity	**Euler VII (Phase 1)**	**Euler VII (Phase 2)**
P_{peak} [GFlops/s]	83.2	78.4
B (STREAM Triad) [GB/s]	19.6681	17.4596
$I_{\text{ridge}} = P_{\text{peak}}/B$ [Flops/Byte]	4.23	4.49

Table 4: Inputs and ridge point of the single-core roofline models.

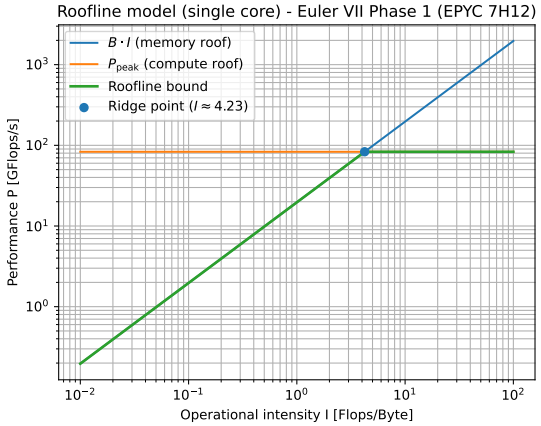


Figure 3: Naive roofline model for a single core on Euler VII (phase 1). The ridge point is at $I_{\text{ridge}} \approx 4.23$ Flops/Byte.

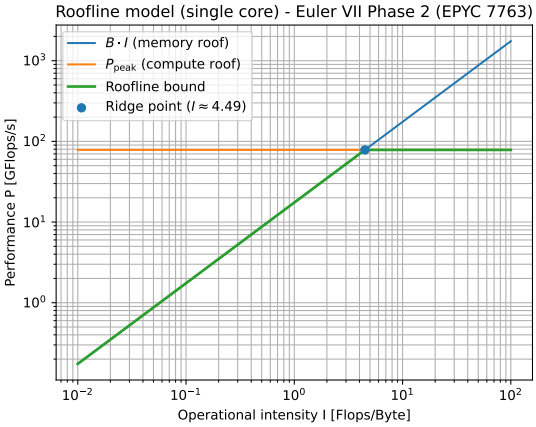


Figure 4: Naive roofline model for a single core on Euler VII (phase 2). The ridge point is at $I_{\text{ridge}} \approx 4.49$ Flops/Byte.

3. Auto-vectorization [10 points]

Question 1: what is data alignment and is it important for data structures to be aligned?

Question 2: what are some obstacles that can prevent automatic vectorization by the compiler?

Question 3: is there a way to help the compiler to vectorize and how?

Question 4: which loop optimizations are performed by the compiler to vectorize and pipeline loops?

Question 5: what can be done if automatic vectorization by the compiler fails or is suboptimal?

4. Matrix multiplication optimization [30 points]

Task 1: Improve the loop ordering in `dgemm_jki.c`.
The original naive implementation uses the loop order i–j–k:

```
void square_dgemm(int n, double* A, double* B, double* C) {
  for (int i = 0; i < n; ++i) {
    for (int j = 0; j < n; ++j) {
      for(int k = 0; k < n; ++k) {
        C[i+j*n] += A[i+k*n] * B[k+j*n];
      }
    }
  }
}
```

Although correct, this ordering is not cache-friendly for column-major storage (indexing $i+jn$). In particular, in the innermost loop over k the access to $A[i+kn]$ jumps across different columns of A with stride n, limiting spatial locality.

To improve data locality, we reorder the loops to j–k–i and hoist invariant quantities out of the innermost loop:

```
void square_dgemm(int n, double* A, double* B, double* C) {
  for (int j = 0; j < n; ++j) {
    double* Ccol = C + j*n; // C(:,j)
    for (int k = 0; k < n; ++k) {
      const double bkj = B[k + j*n]; // B(k,j) reused for all i
      const double* Acol = A + k*n; // A(:,k)
      for (int i = 0; i < n; ++i) {
        Ccol[i] += Acol[i] * bkj; // stride-one on C and A
      }
    }
  }
}
```

With column-major layout, full columns are contiguous in memory. Therefore, for fixed (j, k) the innermost loop over i accesses both $A(:, k)$ and $C(:, j)$ with *stride 1* (contiguous memory), improving cache utilization and hardware prefetching, and enabling better SIMD vectorization. Additionally, loading $B(k, j)$ once into a scalar (`bkj`) and using column pointers (`Acol`, `Ccol`) reduces repeated address computations and memory loads. The computation remains mathematically identical, since it still accumulates

$$C_{i,j} \mathrel{+}= \sum_{k=0}^{n-1} A_{i,k} B_{k,j},$$

but with a more efficient memory-access pattern.

Task 2: Implement blocking for square matrix multiplication in `URLdgemm_blocked.c`
To further increase performance, we implement *blocking* for the matrix multiplication. Instead of multiplying the full matrices at once, we split the iteration space into smaller $BS \times BS$ tiles and compute partial products tile-by-tile:

```
void square_dgemm(int n, double* A, double* B, double* C)
{
  for (int jj = 0; jj < n; jj += BS) {
    int jmax = imin(jj + BS, n);

    for (int kk = 0; kk < n; kk += BS) {
      int kmax = imin(kk + BS, n);

      for (int ii = 0; ii < n; ii += BS) {
        int imax = imin(ii + BS, n);

        for (int j = jj; j < jmax; ++j) {
```

```
        for (int k = kk; k < kmax; ++k) {
          double bkj = B[k + j * n];
          for (int i = ii; i < imax; ++i) {
            C[i + j * n] += A[i + k * n] * bkj;
          }
        }
      }
    }
  }
}
```

The outer loops `jj`, `kk`, and `ii` iterate over tiles of columns/rows with step size `BS`. For each block triple, the inner loops perform the same j–k–i update as in the reordered implementation, but restricted to the current subranges:

$$j \in [jj, j_{\max}), \quad k \in [kk, k_{\max}), \quad i \in [ii, i_{\max}).$$

The helper `imin` ensures correct handling of boundary tiles when n is not a multiple of BS.

Blocking improves cache efficiency because, for a fixed (jj, kk, ii) triple, the algorithm repeatedly reuses the same tiles of A and B while updating the corresponding tile of C. This increases *temporal locality* and reduces the number of times data is fetched from main memory. Inside each tile, the innermost loop still runs over i, giving stride-1 (contiguous) access for $A(:, k)$ and $C(:, j)$ in column-major format. Finally, loading `bkj = B(k,j)` once per (j, k) pair (scalar reuse) avoids redundant memory loads within the innermost loop. We set the default block size to $BS = 64$, which is a commonly used baseline for tiled DGEMM. Blocking aims to improve temporal locality by reusing submatrices (tiles) of A and B while updating a tile of C before moving to the next region. With double-precision elements, a 64×64 tile corresponds to $64^2 \cdot 8 \approx 32\,\text{KiB}$ per matrix tile; the combined working set of the three tiles (A, B, C) is about $3 \cdot 32 \approx 96\,\text{KiB}$, which is small enough to benefit from cache reuse while being large enough to amortize loop overhead. Moreover, using a power-of-two block size simplifies addressing and often interacts well with cache-line behavior. The block size is kept configurable via `#ifndef BS` so that alternative values (e.g. 32 or 48) can be explored if further tuning is desired.